

36. Watch Recognition using OpenCV

Aim

To recognize a watch from a given image using OpenCV.

Procedure

1. Read the input image and watch template image.
2. Convert both images into grayscale.
3. Apply template matching using OpenCV.
4. Find the best matching location.
5. Draw a rectangle around the detected watch.
6. Display the output image.

Code

```
import cv2

# Read images
img = cv2.imread("image.jpg")
template = cv2.imread("watch.jpg")

if img is None or template is None:
    print("Image not found")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
template_gray = cv2.cvtColor(template, cv2.COLOR_BGR2GRAY)

# Template matching
result = cv2.matchTemplate(
    gray,
```

```
    template_gray,  
    cv2.TM_CCOEFF_NORMED  
)  
  
# Find best match  
_, max_val, _, max_loc = cv2.minMaxLoc(result)  
  
# Template size  
h, w = template_gray.shape  
  
# Check detection  
if max_val > 0.5:  
  
    x, y = max_loc  
  
# Draw rectangle  
cv2.rectangle(  
    img,  
    (x, y),  
    (x + w, y + h),  
    (0, 255, 0),  
    2  
)  
  
cv2.putText(  
    img,  
    "Watch",
```

```
(x, y - 10),  
cv2.FONT_HERSHEY_SIMPLEX,  
0.8,  
(0, 255, 0),  
2  
)
```

```
print("Watch detected")  
else:  
    print("Watch not detected")
```

```
print("Matching Score:", max_val)
```

```
# Display output  
cv2.imshow("Watch Recognition", img)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Output



Result

The watch in the given image was successfully recognized and highlighted using OpenCV.